

- 12 (a) signal becomes distorted / noisy B1
 signal loses power / energy / intensity / is attenuated B1 [2]
- (b) (i) *either* numbers involved are smaller / more manageable / cover wider range
or calculations involve addition & subtraction rather than multiplication and division B1 [1]
- (ii) $25 = 10 \lg(P_{\min} / (6.1 \times 10^{-19}))$ C1
 minimum signal power = $1.93 \times 10^{-16} \text{ W}$ C1
 signal loss = $10 \lg(6.5 \times 10^{-3} / (1.93 \times 10^{-16}))$
 = 135 dB C1
 maximum cable length = $135 / 1.6$ C1
 = 85 km so no repeaters necessary A1 [5]